University of Information Technology & Sciences

Assignment – 2
(intermediate level)

Department	: CSE
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Submitted to :

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Designation: Lecturer

Department: CSE

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Sales table.

sale_id	product_id	quantity_sold	sale_date	total_price
1	101	5	2024-01-01	2500.00
2	102	3	2024-01-02	900.00
3	103	2	2024-01-02	60.00
4	104	4	2024-01-03	80.00
5	105	6	2024-01-03	90.00

Sales Table

SQL –

```sql
CREATE TABLE sales(
    sale_id INT PRIMARY KEY,
    product_id INT,
    quantity_sold INT NOT NULL,
    sale_date DATE NOT NULL,
    total_price DECIMAL(10,2),
    FOREIGN KEY (product_id) REFERENCES product(product_id)
);
```

Products table.

product_id	product_name	category	unit_price
101	Laptop	Electronics	500.00
102	Smartphone	Electronics	300.00
103	Headphones	Electronics	30.00
104	Keyboard	Electronics	20.00
105	Mouse	Electronics	15.00

SQL-

```sql
CREATE TABLE product(
    product_id int PRIMARY KEY,
    product_name VARCHAR(50) NOT NULL,
    category VARCHAR(50) NOT NULL,
    unit_price DECIMAL(10,2)
);
```

1. **Calculate the total quantity_sold of products in the 'Electronics' category.**

SQL-

```
SELECT SUM(s.quantity_sold) AS total_quantity_sold
FROM Sales s
JOIN Products p ON s.product_id = p.product_id
WHERE p.category = 'Electronics';
```

Output-

total_quantity_sold
20

2. **Retrieve the product_name and total_price from the Sales table, calculating the total_price as quantity_sold multiplied by unit_price.**

SQL-

```
SELECT p.product_name,(s.quantity_sold * p.unit_price) AS total_price
FROM Sales s
JOIN Products p ON s.product_id = p.product_id;
```

Output-

product_name	total_price
Laptop	2500.00
Smartphone	900.00
Headphones	60.00
Keyboard	80.00
Mouse	90.00

3. Identify the Most Frequently Sold Product from Sales table.

SQL-

```sql
SELECT p.product_name,s.quantity_sold AS total_sold
FROM Sales s
JOIN Products p ON s.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_sold DESC
LIMIT 1;
```

Output-

product_name	total_sold
Mouse	6

4. Find the Products Not Sold from Products table.

SQL-

```sql
SELECT p.product_name
FROM Products p
LEFT JOIN Sales s ON p.product_id = s.product_id
WHERE s.product_id IS NULL;
```

Output-

product_name

5. Calculate the total revenue generated from sales for each product category.

SQL-

```sql
SELECT p.category,SUM(s.quantity_sold * p.unit_price) AS total_revenue
FROM Sales s
JOIN Products p ON s.product_id = p.product_id
GROUP BY p.category;
```

Output-

category	total_revenue
Electronics	3630.00

6. Find the product category with the highest average unit price.

SQL-

```sql
SELECT category,AVG(unit_price) AS avg_unit_price
FROM Products
GROUP BY category
ORDER BY avg_unit_price DESC
LIMIT 1;
```

Output-

←T→	category	avg_unit_price
Edit Copy Delete	Electronics	173.000000

7. Identify products with total sales exceeding 30.

SQL-

```sql
SELECT p.product_name,SUM(s.quantity_sold) AS total_sales
FROM Sales s
JOIN Products p ON s.product_id = p.product_id
GROUP BY p.product_name
HAVING SUM(s.quantity_sold) > 30;
```

Output-

product_name	total_sales

8. Determine the average quantity sold for products with a unit price greater than $100.

SQL-

```sql
SELECT AVG(s.quantity_sold) AS avg_quantity
FROM Sales s
JOIN Products p ON s.product_id = p.product_id
WHERE p.unit_price > 100;
```

Output-

avg_quantity
4.0000

9. Retrieve the product name and total sales revenue for each product.

SQL-

```sql
SELECT p.product_name,SUM(s.quantity_sold * p.unit_price) AS
total_revenue
FROM Sales s
JOIN Products p ON s.product_id = p.product_id
GROUP BY p.product_name;
```

Output-

product_name	total_revenue
Headphones	60.00
Keyboard	80.00
Laptop	2500.00
Mouse	90.00
Smartphone	900.00

10. List all sales along with the corresponding product names.

SQL-

```
SELECT s.*, p.product_name
FROM Sales s
JOIN Products p ON s.product_id = p.product_id;
```

Output-

sale_id	product_id	quantity_sold	sale_date	total_price	product_name
1	101	5	2024-01-01	2500.00	Laptop
2	102	3	2024-01-02	900.00	Smartphone
3	103	2	2024-01-02	60.00	Headphones
4	104	4	2024-01-03	80.00	Keyboard
5	105	6	2024-01-03	90.00	Mouse

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